



Gayathri K 251801076-2025-AIDS ▾

G2**Started on** Thursday, 27 August 2026, 10:30 AM**State** Finished**Completed on** Thursday, 27 August 2026, 10:31 AM**Time taken** 43 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int findFirstZero(int arr[], int low, int high)
4  {
5      if (low > high)
6          return -1;
7
8      int mid = low + (high - low) / 2;
9
10     if (arr[mid] == 0)
11     {
12         if (mid == 0 || arr[mid - 1] == 1)
13             return mid;
14
15         return findFirstZero(arr, low, mid - 1);
16     }
17     else
18     {
19         return findFirstZero(arr, mid + 1, high);
20     }
21 }
22
23 int main()
24 {
25     int n;
26     scanf("%d", &n);
27
28     int arr[n];
29
30     for (int i = 0; i < n; i++)
31         scanf("%d", &arr[i]);
32
33     int firstZero = findFirstZero(arr, 0, n - 1);
34
35     if (firstZero == -1)
36         printf("0");
37     else
38         printf("%d", n - firstZero);
39
40     return 0;
41 }
42

```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Gayathri K 251801076-2025-AIDS ▾

G2

Started on	Thursday, 27 August 2026, 10:31 AM
State	Finished
Completed on	Thursday, 27 August 2026, 10:32 AM
Time taken	33 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countElement(int arr[], int left, int right, int value)
4  {
5      int count = 0;
6
7      for (int i = left; i <= right; i++)
8      {
9          if (arr[i] == value)
10             count++;
11      }
12
13      return count;
14  }
15
16  int majorityElement(int arr[], int left, int right)
17  {
18      if (left == right)
19          return arr[left];
20
21      int mid = left + (right - left) / 2;
22
23      int leftMajority = majorityElement(arr, left, mid);
24      int rightMajority = majorityElement(arr, mid + 1, right);
25
26      if (leftMajority == rightMajority)
27          return leftMajority;
28
29      int leftCount = countElement(arr, left, right, leftMajority);
30      int rightCount = countElement(arr, left, right, rightMajority);
31
32      if (leftCount > rightCount)
33          return leftMajority;
34      else

```

```
35         return rightMajority;
36     }
37
38     int main()
39     {
40         int n;
41         scanf("%d", &n);
42
43         int arr[n];
44
45         for (int i = 0; i < n; i++)
46             scanf("%d", &arr[i]);
47
48         printf("%d", majorityElement(arr, 0, n - 1));
49
50         return 0;
51     }
52
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Gayathri K 251801076-2025-AIDS ▾

G2

Started on	Thursday, 27 August 2026, 10:32 AM
State	Finished
Completed on	Thursday, 27 August 2026, 10:33 AM
Time taken	42 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findFloor(int arr[], int low, int high, int x)
4  {
5      if (low > high)
6          return -1;
7
8      int mid = low + (high - low) / 2;
9
10     if (arr[mid] == x)
11         return arr[mid];
12
13     if (arr[mid] < x)
14     {
15         int result = findFloor(arr, mid + 1, high, x);
16
17         if (result != -1)
18             return result;
19
20         return arr[mid];
21     }
22
23     return findFloor(arr, low, mid - 1, x);
24 }
25
26 int main()
27 {
28     int n;
29     scanf("%d", &n);
30
31     int arr[n];
32
33     for (int i = 0; i < n; i++)
34         scanf("%d", &arr[i]);
35
36     int x;
37     scanf("%d", &x);
38
39     int result = findFloor(arr, 0, n - 1, x);
40
41     printf("%d", result);
42
43     return 0;
44 }
45
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Started on	Thursday, 27 August 2026, 10:33 AM
State	Finished
Completed on	Thursday, 27 August 2026, 10:34 AM
Time taken	34 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findPair(int arr[], int left, int right, int x)
4  {
5      if (left >= right)
6          return 0;
7
8      int sum = arr[left] + arr[right];
9
10     if (sum == x)
11     {
12         printf("%d\n", arr[left]);
13         printf("%d\n", arr[right]);
14         return 1;
15     }
16
17     if (sum < x)
18         return findPair(arr, left + 1, right, x);
19
20     return findPair(arr, left, right - 1, x);
21 }
22
23 int main()
24 {
25     int n;
26     scanf("%d", &n);
27
28     int arr[n];
29
30     for (int i = 0; i < n; i++)
31         scanf("%d", &arr[i]);
32
33     int x;
34     scanf("%d", &x);
35
36     if (!findPair(arr, 0, n - 1, x))
37         printf("No");
38
39     return 0;
40 }
41
```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Back to Course



Gayathri K 251801076-2025-AIDS ▾

G2**Started on** Thursday, 27 August 2026, 10:34 AM**State** Finished**Completed on** Thursday, 27 August 2026, 10:35 AM**Time taken** 48 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5	12 34 67 78 98
67 34 12 98 78	

Answer:

```

1  #include <stdio.h>
2
3  void swap(int *a, int *b)
4  {
5      int temp = *a;
6      *a = *b;
7      *b = temp;
8  }
9
10 int partition(int arr[], int low, int high)
11 {
12     int pivot = arr[high];
13     int i = low - 1;
14
15     for (int j = low; j < high; j++)
16     {
17         if (arr[j] <= pivot)
18         {
19             i++;
20             swap(&arr[i], &arr[j]);
21         }
22     }
23
24     swap(&arr[i + 1], &arr[high]);
25
26     return i + 1;
27 }
28
29 void quickSort(int arr[], int low, int high)
30 {
31     if (low < high)
32     {
33         int pi = partition(arr, low, high);
34
35         quickSort(arr, low, pi - 1);
36         quickSort(arr, pi + 1, high);
37     }
38 }
39
40 int main()
41 {
42     int n;
43     scanf("%d", &n);
44
45     int arr[n];
46
47     for (int i = 0; i < n; i++)
48         scanf("%d", &arr[i]);
49
50     quickSort(arr, 0, n - 1);
51
52     for (int i = 0; i < n; i++)

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)